# POCKETSMART AI

Your Smart Budget & Recommendation Assistant

COLLEGE PROJECT DOCUMENT

# 1. ABSTRACT

PocketSmart AI is an intelligent personal finance assistant designed to help users manage their money effectively. The system allows users to record income and expenses, create budgets, monitor spending habits, and receive personalized financial recommendations. Using Artificial Intelligence, PocketSmart AI analyzes spending patterns and provides suggestions for reducing unnecessary expenses and improving savings.

# 2. INTRODUCTION

Managing personal finances can be difficult because people often do not track daily expenses or understand where their money is being spent. PocketSmart AI provides a simple digital solution for managing personal budgets. The application collects information about income, expenses, savings, and spending categories and provides useful insights through AI-based recommendations.

# 3. PROBLEM STATEMENT

Many people face difficulties in tracking daily expenses, maintaining monthly budgets, identifying unnecessary spending, planning savings, understanding spending patterns, and making personalized financial decisions. PocketSmart AI aims to solve these problems through an easy-to-use intelligent budgeting assistant.

# 4. OBJECTIVES

- Provide an easy platform for recording income and expenses.
- Help users create and maintain budgets.
- Categorize expenses automatically or manually.
- Analyze spending patterns using AI.
- Provide personalized recommendations.
- Monitor savings and budget limits.
- Generate useful financial reports.

# 5. PROPOSED SYSTEM

The proposed system is a web/mobile-based personal finance application. Users enter income and expenses into the system. The application stores and analyzes the data. The AI recommendation component examines spending behavior and provides suggestions such as reducing unnecessary purchases, controlling category-wise spending, and increasing savings.

# 6. KEY FEATURES

- User registration and login
- Income management
- Expense tracking
- Expense categorization
- Monthly budget creation
- Savings tracking
- Spending analysis

• AI-based recommendations
• Budget alerts
• Dashboard and reports
• Transaction history

# 7. SYSTEM MODULES

**User Management:** Users can register, log in, and manage profiles securely.

**Income Management:** Users can enter salary, allowance, freelance income, or other income sources.

**Expense Management:** Users can record expenses such as food, transportation, education, shopping, bills, and entertainment.

**Budget Management:** Users can set monthly or category-based spending limits.

**AI Recommendation:** AI analyzes transaction history and identifies spending patterns to provide personalized recommendations.

**Dashboard:** Displays total income, expenses, savings, remaining budget, and spending categories.

**Reports:** Provides daily, weekly, and monthly financial summaries.

# 8. SYSTEM ARCHITECTURE

User → User Interface → Application Server → Database

The application server processes user transactions and sends the required information to the AI recommendation engine.

AI Recommendation Engine → Spending Analysis → Personalized Recommendation → User

# 9. TECHNOLOGIES USED

**Frontend:** HTML, CSS, JavaScript
**Backend:** Python / Node.js
**Database:** MySQL / SQLite
**AI/ML:** Python, Machine Learning algorithms, recommendation techniques

# 10. AI RECOMMENDATION PROCESS

1. Collect user transaction data.
2. Categorize income and expenses.
3. Calculate spending by category.
4. Identify unusual or excessive spending.

5. Analyze historical spending patterns.
6. Compare spending with the user's budget.
7. Generate personalized recommendations.
8. Display recommendations on the dashboard.

## 11. EXAMPLE RECOMMENDATION

If a user spends a large percentage of monthly income on food delivery, PocketSmart AI can identify the pattern and provide a suggestion: “Your food expenses are higher than your planned budget this month. Consider reducing food-delivery spending to improve your monthly savings.”

## 12. DATABASE DESIGN

**User Table:** User_ID, Name, Email, Password
**Income Table:** Income_ID, User_ID, Amount, Source, Date
**Expense Table:** Expense_ID, User_ID, Category, Amount, Description, Date
**Budget Table:** Budget_ID, User_ID, Category, Limit, Month
**Recommendation Table:** Recommendation_ID, User_ID, Message, Date

## 13. ADVANTAGES

• Easy expense tracking.
• Helps users control spending.
• Saves time through automated analysis.
• Provides personalized recommendations.
• Helps improve budgeting habits.
• Provides clear financial reports.
• Can be used anytime and anywhere.

## 14. FUTURE ENHANCEMENTS

• Voice-based expense entry.
• Receipt scanning using OCR.
• Advanced AI financial insights.
• Bank account integration.
• Investment recommendations.
• Bill-payment reminders.
• Predictive monthly expense analysis.
• Multi-language support.
• Mobile application support.

## 15. CONCLUSION

PocketSmart AI is an intelligent budgeting and recommendation system that helps users manage personal finances. By combining expense tracking, budgeting, data analysis, and AI-based recommendations, the system provides useful insights into financial habits. The project can help

users become more aware of spending, maintain budgets, and work toward better saving habits.

# 16. REFERENCES

1. Python Documentation
2. MySQL Documentation
3. HTML, CSS and JavaScript Documentation
4. Machine Learning and Recommendation System concepts
5. Personal Finance Management concepts